Mixed-Signal Design Automation and Verification

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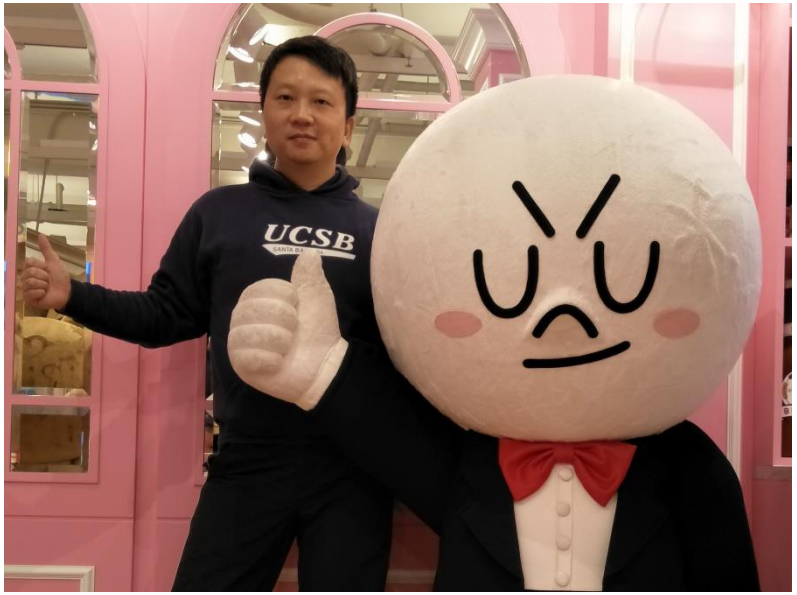
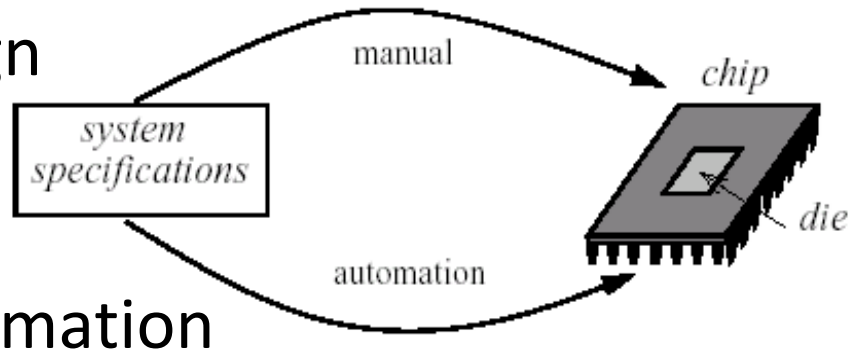
實驗室指導教授

- 姓名：劉建男
- 現職：教授兼系主任
- 專長：VLSI/CAD（電腦輔助設計）
- 經歷：
  - 陽明交通大學教授(2018.8 -)
  - 陽明交通大學電子所副所長
  - 中央大學副教務長
  - 教育部桃竹苗區域教學資源中心主任
  - ACM SIGDA台灣支會主席
  - IEEE CASS台北支會副主席
  - UCLA, UCSB訪問學者
  - 中央大學教授(2001.8-2018.7)



- 榮譽：91學年度優良導師, 92學年度資電院教學優良教師, 94學年度教學傑出獎, 2011 TICD 傑出年輕學者, 104學年度教學優良獎, 108學年度績優導師, …

CAD Research in VLSI Design



- CAD = Computer-Aided Design
  - Exist everywhere in the whole design flow
- EDA = Electronic Design Automation
  - CAD for electronic industry
- Typically, more than 50% design time is elapsed on running CAD tools
  - Important but often invisible
- CAD research falls between design (hardware) and programming (software)
  - Require knowledge from both parts
- Methodology (thinking) + program (implementation)

Why EDA is Important?

EDA 產業則是 IC 設計的最上游產業，也是積體電路設計必需，且最重要的軟體。

EDA 在半導體領域中產值小，但極為重要，數據顯示，2018 年 EDA 的市場規模只有 97.15 億美元（約新台幣 2914.5 億新台幣），相對於幾千億美金的半導體產業產值而言貢獻度極低。但如果缺少這個產業，全球的 IC 設計公司都將停擺。

TechOrange 科技報橘

徵才 數位分身最強應用 台灣資安新知 智慧製造 創新嗜讀 CXO 管理學 資安

失去它華為就癱瘓！帶你認識美國的制裁王牌：EDA 產業

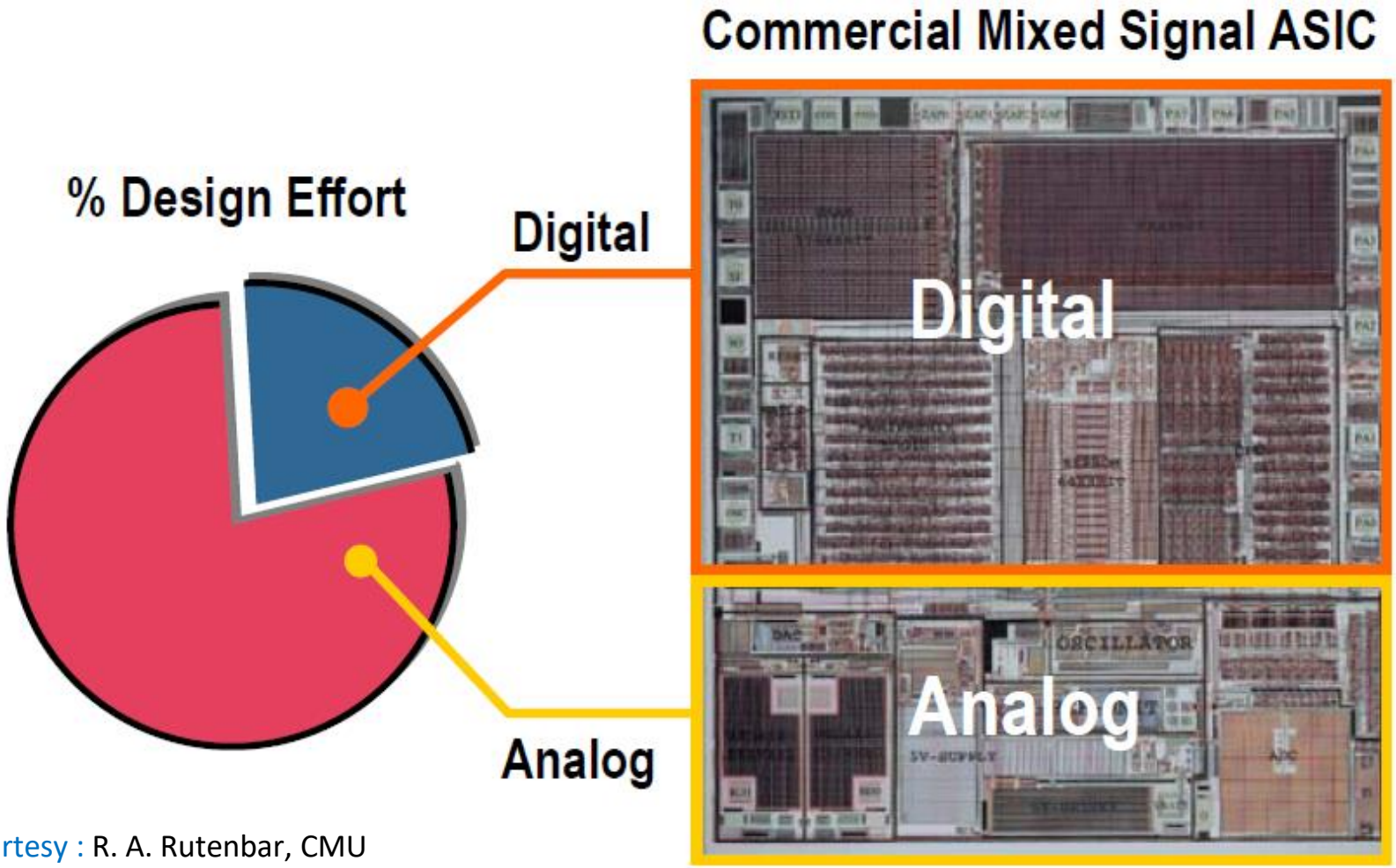
2020/05/20 讚 6,409 分享

鉅亨網

【為什麼我們要挑選這篇文章】美國加強封殺華為，除了晶圓代工之外，「EDA」也是美國手中握有的王牌。EDA 是電子設計自動化，是半導體製造業中最上游的產業，雖然產值不高，但如果停擺，下游的設計、製造、封裝皆無法運作。由於美國 EDA 廠持有關鍵技術，若暫停與華為合作，華為最糟可能跟中興一樣，連產品都無法銷售。（責任編輯：郭家宏）

https://buzzorange.com/techorange/2020/05/20/eda-is-important-for-huawei/

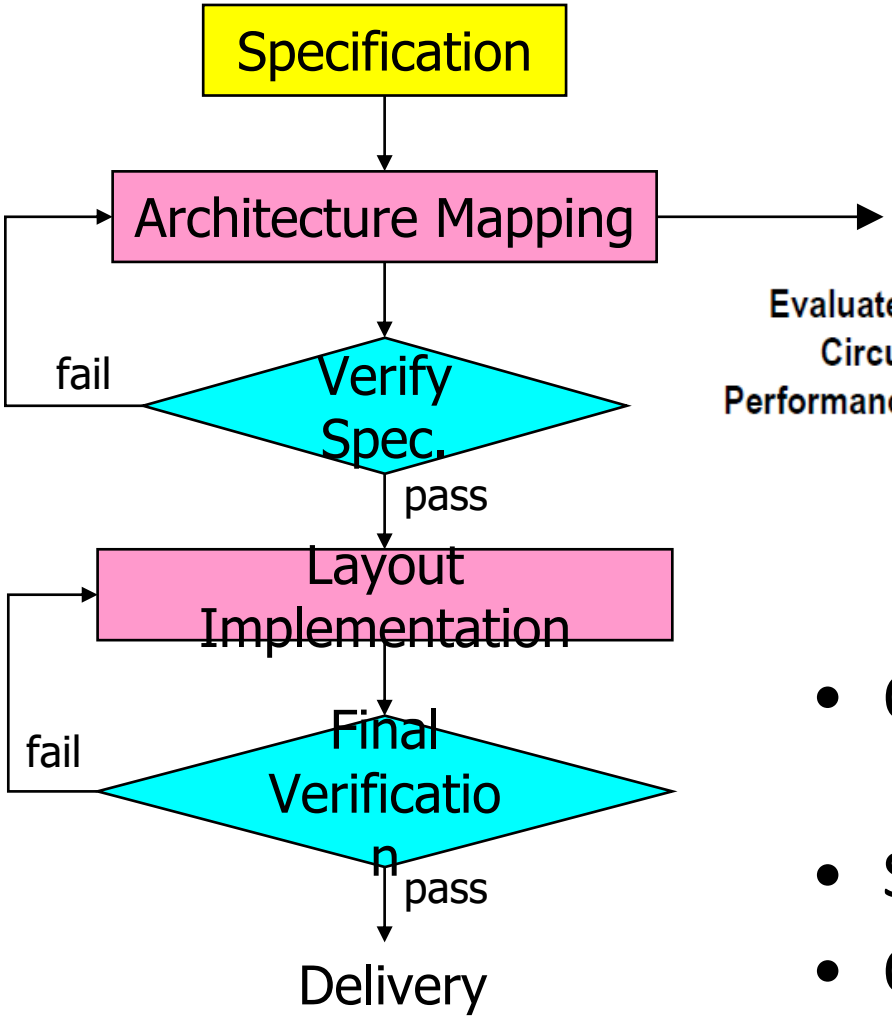
The Mixed-Signal Design Problem

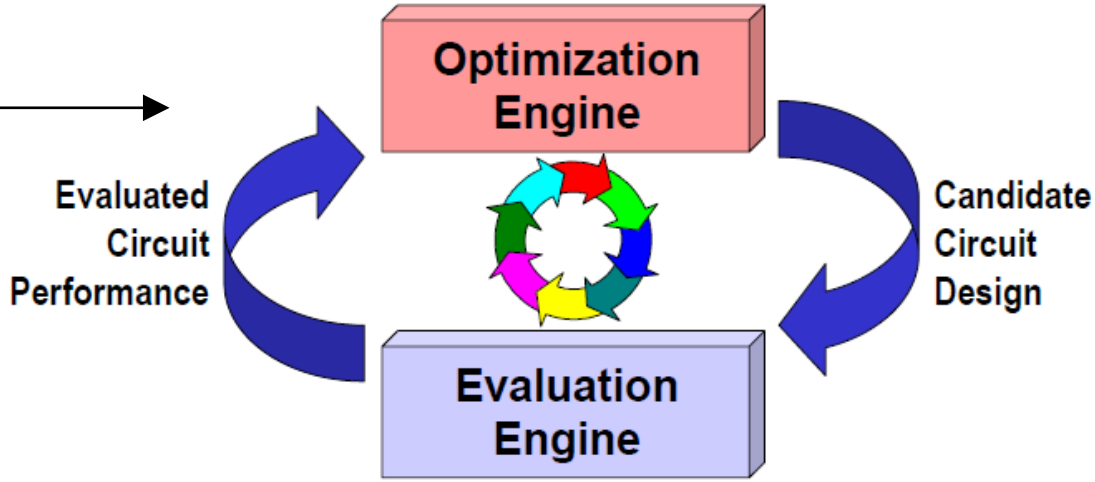


Courtesy : R. A. Rutenbar, CMU

WHY?

Designing Analog Circuits is not Easy !!

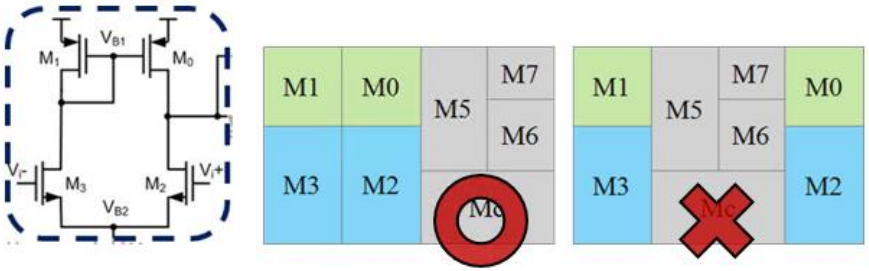


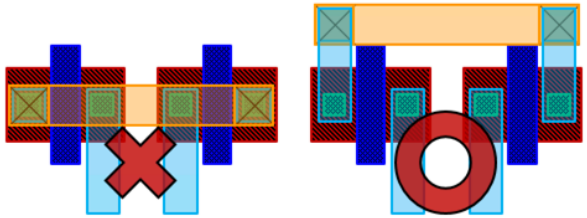


- Often require many iterations
  - Hard to meet all spec.
- Simulation time is long
- Optimality is not guaranteed

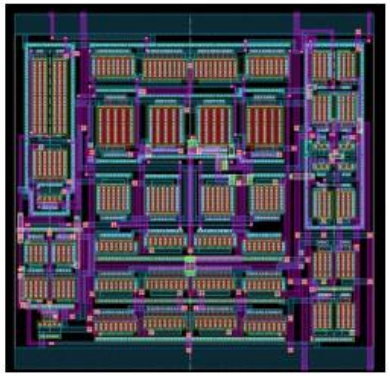
Analog Layout Design still Done Manually

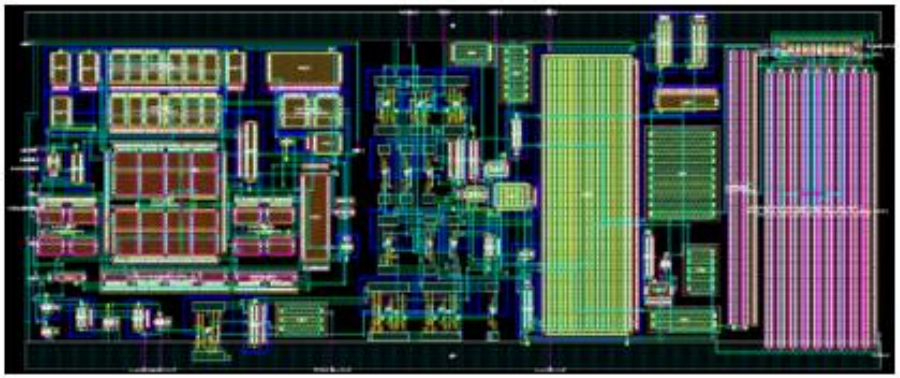
- Take too much time to consider more and more layout effects for analog circuits, ex:
  - Topology constraints for placement
  - Avoid over-the-cell routing net





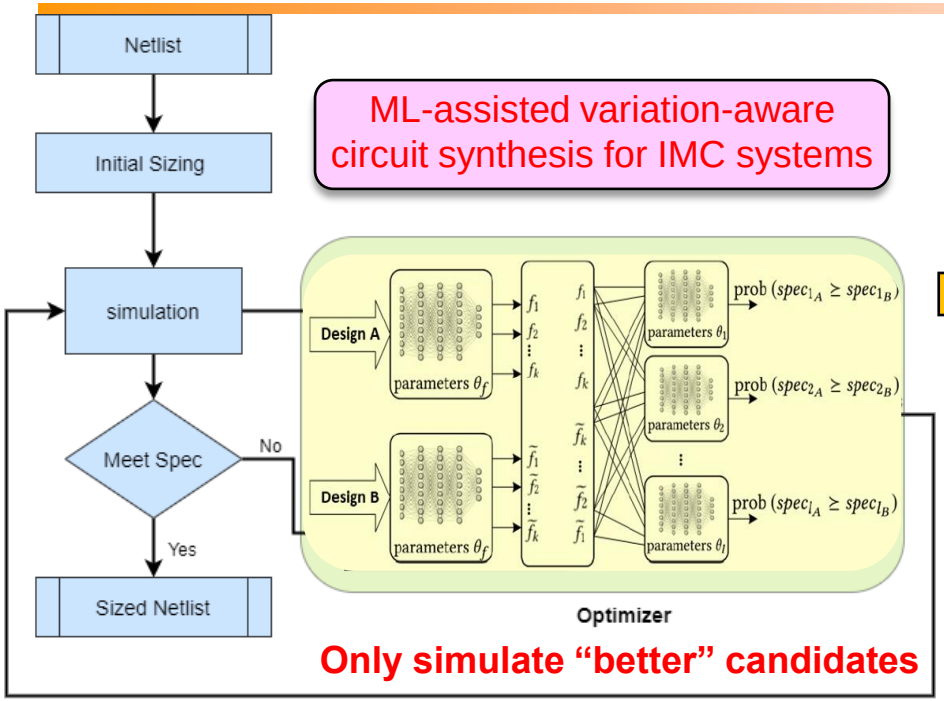
- Each designer has his own layout "style" ⇒ experience

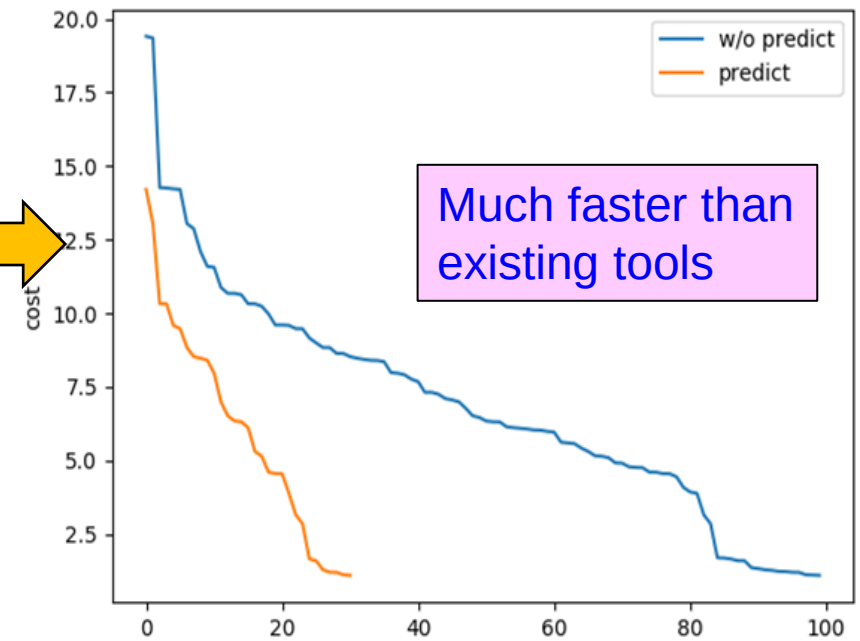




Source: Cadence

DNN-Assisted Analog Circuit Sizing



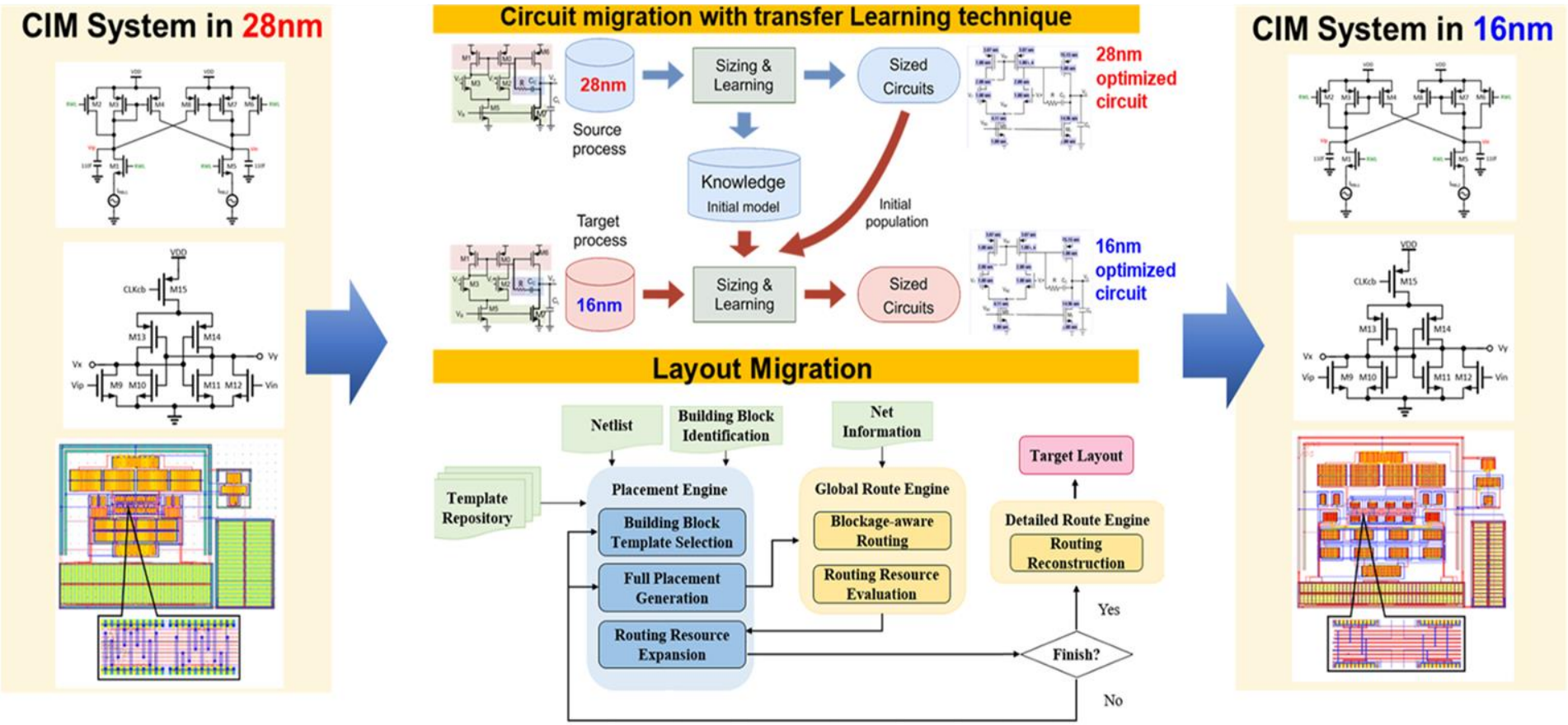


	Spec.	Gain [db]	PM [deg]	# sim.	Speedup
		> 70	> 80	-	-
ADC input buffer (TSMC 28nm)	Traditional EA	70.01	86.64	1990	1 x
	ML-assisted EA	70.01	85.74	320	6.18 x
	Parallel 2 process	70.02	82.74	342	10.21 x
	Parallel 4 process	70.01	83.21	357	20.06 x

Sense Amplifier (TSMC 28nm)	Manual sizing	ML-Assisted EA (4 proc.)	Improve-ment
Yield (MC 1000)	71.8%	99.90%	+28.1%
Power	103.3 µW	26.96 µW	-73.9%
Area	15.98 µm²	11.54 µm²	-27.8%
Sizing time	several days	54.3 min	a lot

Automatic Analog Design Migration

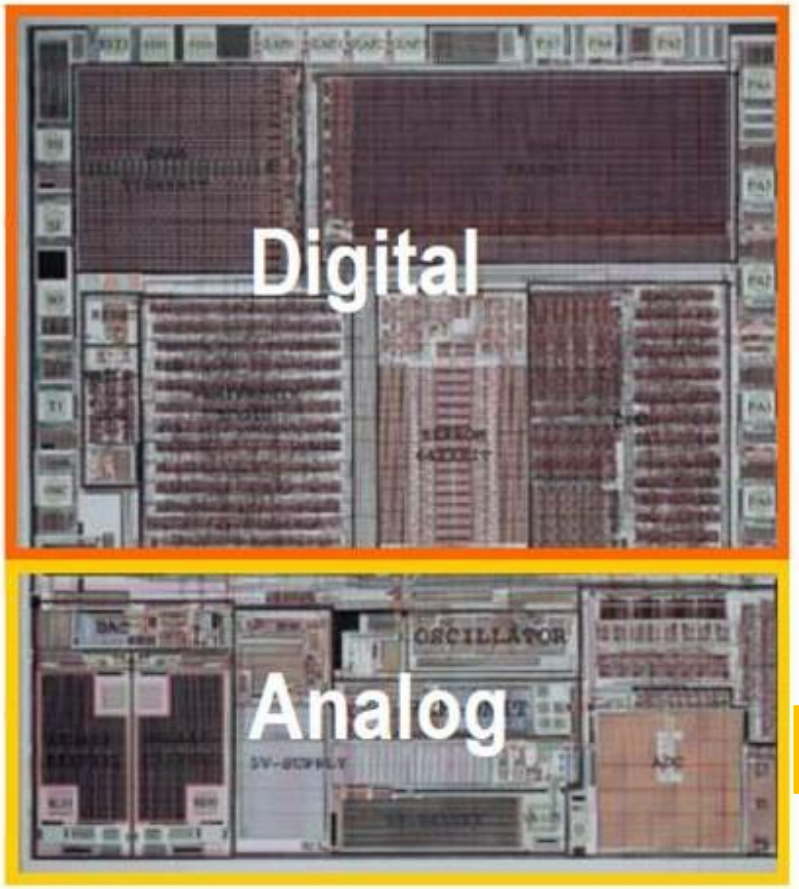
- Refer to the original design in old technology (ex:28nm), automatically generate the design & layout in new tech.



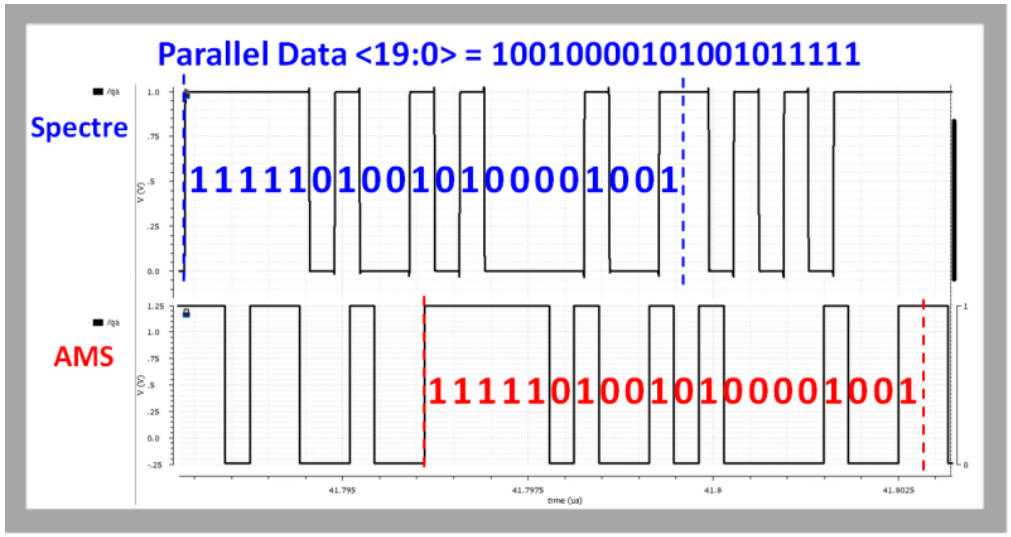
Analog Behavior Modeling for System Verification

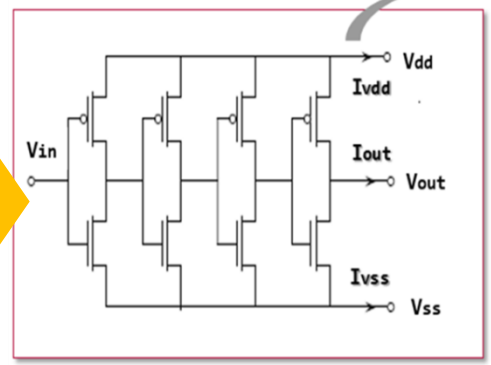
- Automatically build up behavioral models for analog circuits
  - Fast & accurate for system simulation with digital circuits

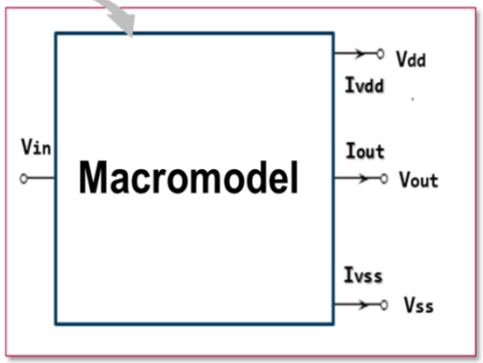
Commercial Mixed Signal ASIC



sim. time: 96:45:00 v.s 00:39:25 (147x)

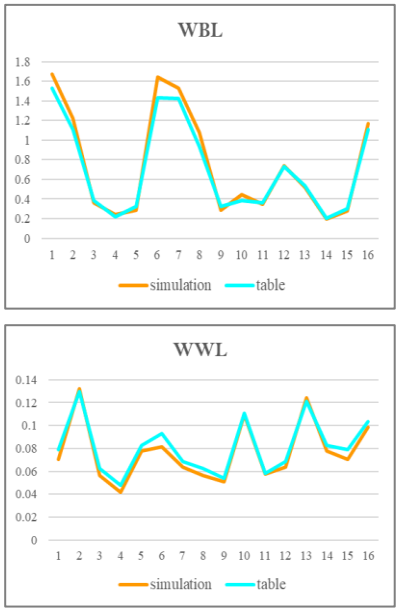




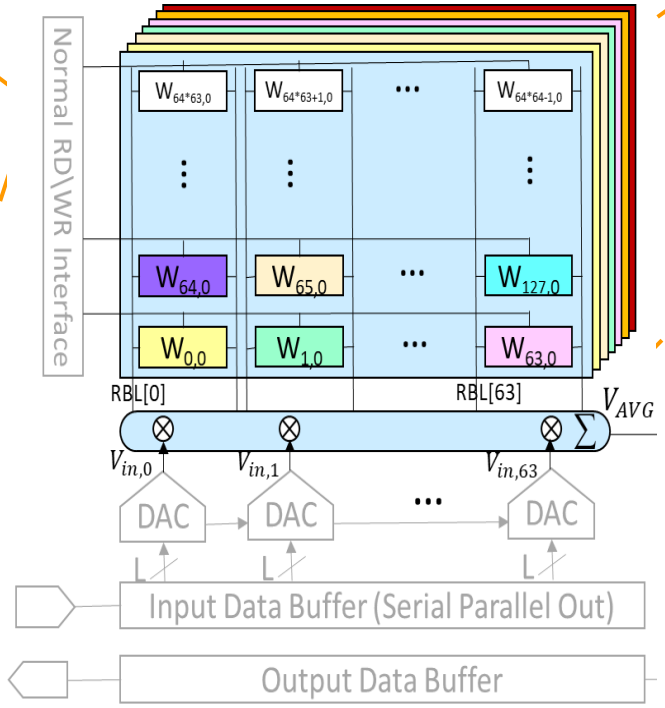


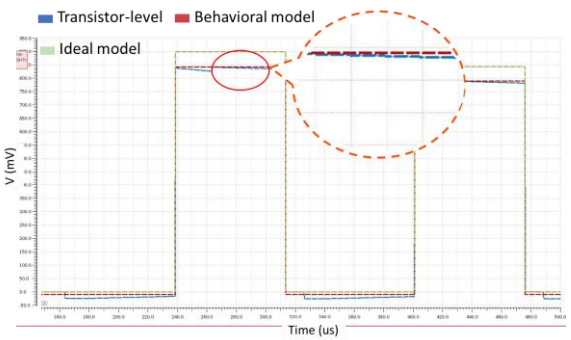
System Behavioral Models for IMC

- Dynamic **power** model for IMC memory array
  - System-level power calculation for applications
- System-level behavioral models
  - Coupling-aware model for cell array
  - Non-ideal model for digital-to-analog converter (DAC)

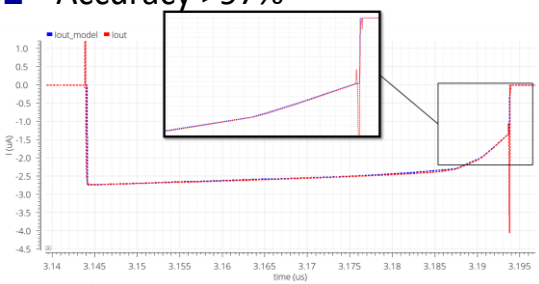


- Speed up 100x
- Accuracy > 94%



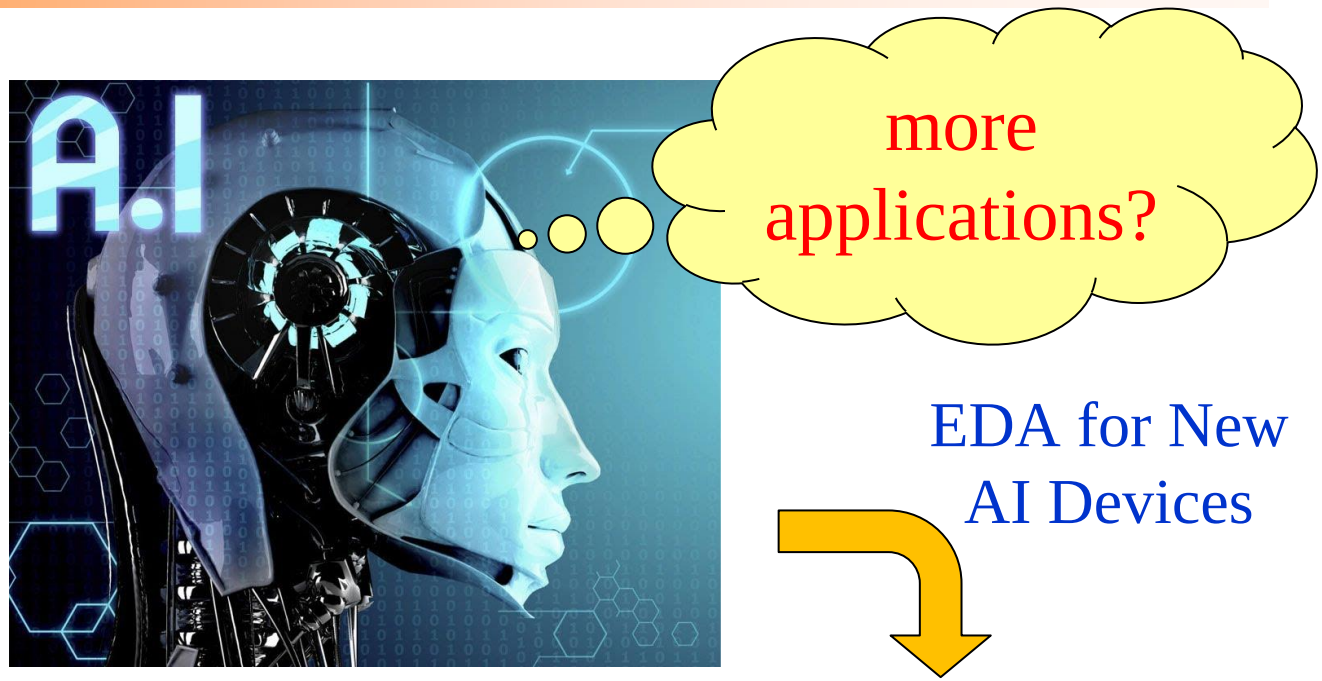


- Speed up 300x
- Accuracy >97%



- Speed up 260x
- Accuracy >87%

Future Projects: AI + EDA



AI for EDA Algorithms

EDA for New AI Devices

Lab: ED-413

MSEDA

Lab Introduction

Web Site